



Foundations of Software Engineering & The Software Crisis

Lecture 01: Systems Evolution, Catastrophic Failures, and Academic Taxonomy

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The Dual Role & Socio-Technical Nature of Software

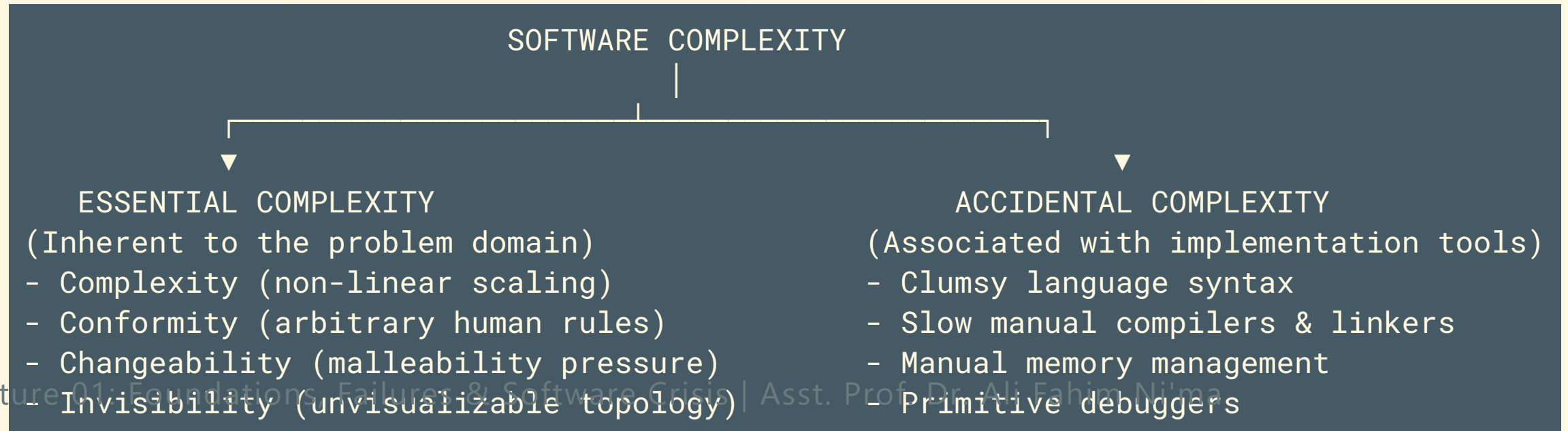
- **Dual Essence of Modern Software:**
 1. **Software as a Product:** Transforms data, executes business logic, and powers cyber-physical devices.
 2. **Software as a Vehicle:** Acts as the operational platform controlling hardware, operating systems, and communications infrastructure.
- **The Socio-Technical Reality (Sommerville):**
 - Software does not operate in an abstract mathematical bubble.
 - A complete system includes **code, operational processes**, and

Case Study: Dhahran Patriot Missile Failure (1991)

- **The Disaster:** On Feb 25, 1991, an Iraqi Scud missile struck US barracks in Dhahran, killing 28 soldiers after the Patriot battery failed to engage.
- **The Mathematical Root Cause:**
 - System clock incremented in integer tenths of a second (0.1 s).
 - Converted to seconds by multiplying by a **24-bit fixed-point register** truncation of 1/10.
 - Truncation error: $\Delta t \approx 0.0000000095$ seconds per tick.

Brooks' Thesis: "No Silver Bullet" (1986)

“ *"There is no single development, in either technology or management technique, which by itself promises even one order of magnitude (10×) improvement in productivity, in reliability, in simplicity, within a decade."* ”



What Software Truly Is: The Tripartite Asset

Software is **not** merely executable code written by a programmer.

Software = Programs + Data Structures + Documentation

1. Programs (Code):

- High-quality, modular, testable, and maintainable source code and build scripts.

2. Data Structures:

- Schemas, state representations, relational and in-memory data representations.



Defining Software Engineering & The Software Process

- **IEEE Standard 610.12 Definition:**

“ *"The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering to software."* ”

- **The Four Universal Process Activities:**

1. **Specification:** Defining system capabilities and operational boundaries.

2. **Development:** Architectural design, decomposition, and

The Software Crisis: Genesis & The IBM OS/360 Lessons

- **Coined at NATO Conference (Garmisch, 1968):**
 - Software projects chronically over budget (200%--500%), years behind schedule, and untrustworthy.
 - Maintenance costs escalating to consume 70%--80% of total IT budgets.
- **The IBM System/360 Experience (Fred Brooks):**
 - Consumed **5,000+ man-years** and over **\$50,000,000**.
 - Suffered years of schedule delays and thousands of latent defect releases.

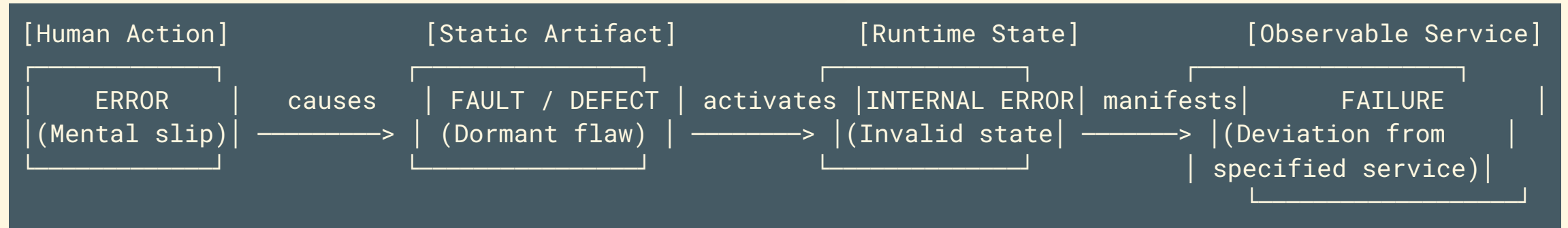


Debunking Software Myths

Category	Persistent Industry Myth	Empirical Software Engineering Reality
Management	<i>"If we fall behind schedule, we can just hire more programmers."</i>	Brooks' Law: Increases communication overhead and training drag, causing further delay.
Customer	<i>"A vague general objective is enough; we can work out details later."</i>	Ambiguous requirements are the #1 cause of software failure and architectural rework.
Developer	<i>"Once the code compiles and runs, our job as engineers is complete."</i>	Post-delivery maintenance accounts for 60%--80% of the entire system lifecycle cost.
Developer	<i>"The working program is the only deliverable that matters."</i>	Code without documentation is a maintenance disaster that guarantees technical bankruptcy.

Core Terminology Taxonomy: Dependability Chain

Under formal IEEE 610.12 and Laprie / Avižienis standards:



- **Error:** Cognitive human mistake or oversight.
- **Fault / Defect:** Static flaw residing silently in code or documentation.
- **Failure:** Dynamic event where delivered service deviates from

Professional Ethics & Open Defense Viva Questions

- **ACM/IEEE Code of Ethics (8 Core Principles):**

1. Public · 2. Client/Employer · 3. Product · 4. Judgment · 5. Management · 6. Profession · 7. Colleagues · 8. Self.

Open Viva Defense Questions (Dr. Ali Fahim):

1. *"How does the Patriot missile failure demonstrate that software maintenance is often more hazardous than initial greenfield development?"*